

LAHORE COLLEGE FOR WOMEN UNIVERSITY, LAHORE

Mid Term Examination 2025

Paper: Computer Architecture

Max Marks: 30

Time Allowed: 1 hour 15 mins

Q1) What is instruction set design. Discuss any two instruction set design with the help of example.

Q2) Describe the Following

a) Difference between RISC and CISC.

b) Flynn's taxonomy and describe any two types

Q3) Assume the stages of pipelined are perfectly balanced (ideal conditions), Calculate the time per instruction on the pipelined processor for both instructions a and b separately.

	Fetch	Decode	Execute	Memory	Writeback
a.	300ps	400ps	350ps	550ps	100ps
b.	200ps	150ps	100ps	190ps	140ps

Q # 4 what is cache memory? Discuss the significance of Cache memory

$$T_{total} = (n-1) \times T_{stage} + T_{writeback}$$

$$= (5-1) \times 100 + 100 = 500ps$$